MACAD 2025/26 · DIGITAL TOOLS FOR GRAPH MACHINE LEARNING

GraphML_G02

Graph Learning Pipeline for Architectural Floor Plans

FINAL ASSIGNMENT

IAAC · MaCAD 2025/26

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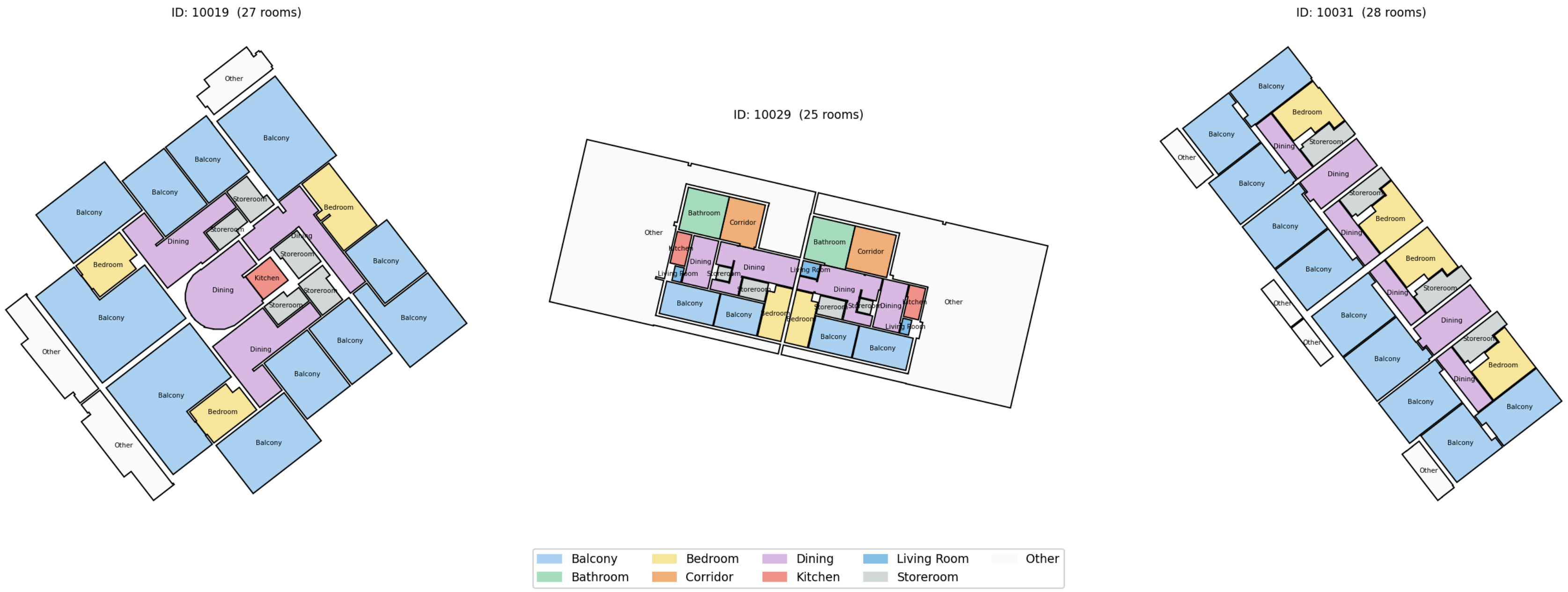
Contents

This week we explored the Swiss Dwellings dataset and the research papers behind it, and defined the next steps for our graph ML pipeline.

01 DATASET

Modified Swiss Dwellings

5,372 floor plans | Multi-apartment clusters | Corridor buildings | Single units | Graph modality only



3 sample plans — wide variety of residential typologies, scales and orientations

5,372	4	9	.pickle
Floor Plans	Zone Types	Room Types	Graph Format

01 DATASET — SELECTED PLAN

Plan 10000 — Multi-Apartment Cluster

Room types — rendered from graph_out geometry

41 **Rooms**
across 2 apartments

2 **Apartments**
sharing a common entrance

✓ **Ground Truth**
validate predictions against MSD labels

Cluster typology **Diagonal layout**

02 RESEARCH

Literature

PAPER 1

MSD: A Benchmark Dataset for Floor Plan Generation of Building Complexes

ECCV 2024 · TU Delft

- 5,372 Swiss residential floor plans
- 3 modalities: image, geometry, graph
- Richer than RPLAN & LIFULL — multi-apartment, compass orientation, inter-unit links
- Graph encodes spatial zones, not room types

> This dataset is our primary data source.

PAPER 2

GNN for Node Classification and Attribute Allocation in Architectural BIM

eCAADe 2024 · Cardiff University

- Uses the same MSD dataset
- Converts floor plans to graphs via TopologicPy
- Benchmarks GCN, GAT, GraphSAGE architectures
- GraphSAGE-Pool, 4 layers — ~95% accuracy

> The pretrained model from this paper is what we will run on our floor plan.

03 GRAPH STRUCTURE

Data Structure

graph_in — model input

Attribute	Type	Values
`zoning_type`	int	**0 · 1 · 2 · 3**
`connectivity`	str	**'door' · 'entrance'**

graph_out — ground truth

Attribute	Type	Description
`room_type`	int	**0 – 8**
`geometry`	Polygon	2D room outline
`centroid`	Point	centre of room

Room types (0 – 8)

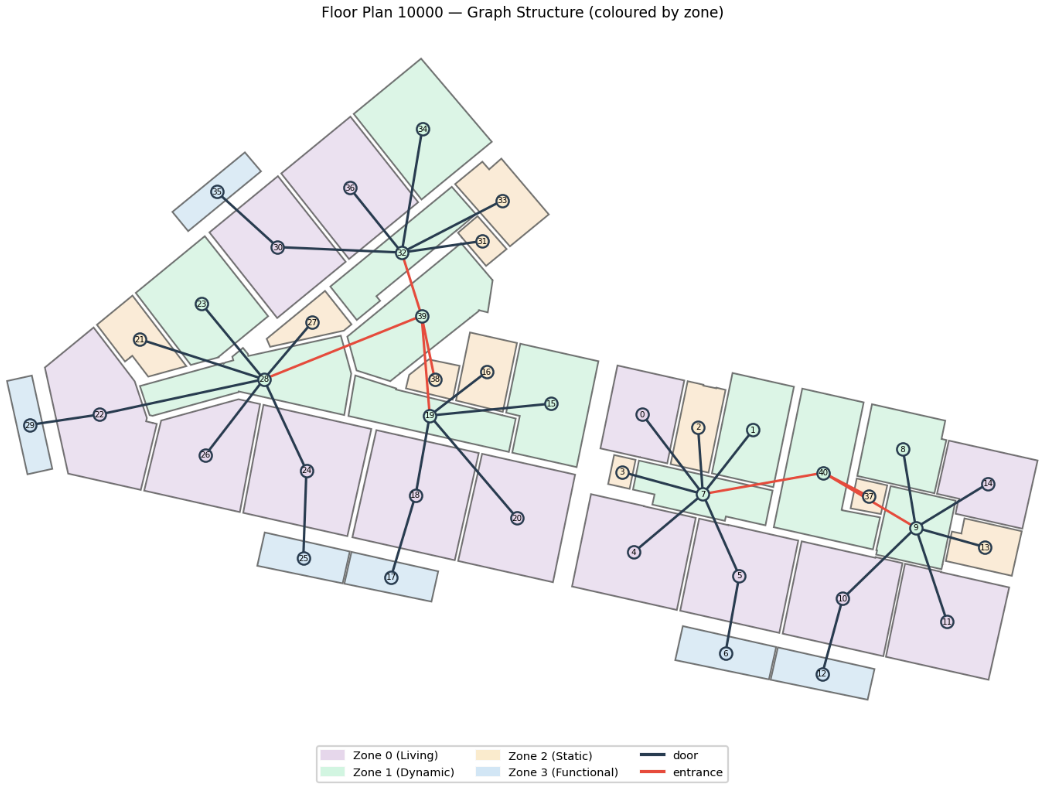
ID	Room	ID	Room
0	Balcony	**5**	Kitchen
1	Bathroom	**6**	Living Room
2	Bedroom	**7**	Storeroom
3	Corridor	**8**	Other
4	Dining		

Zone types (0 – 3)

ID	Zone	Typical rooms
0	Living	Living room, dining
1	Dynamic	Corridor, entrance
2	Static	Bedroom, bathroom
3	Functional	Kitchen, storeroom

03 GRAPH STRUCTURE

From Plan to Graph — Plan 10000



Plan 10000 — room types (left) and spatial connectivity graph (right)

Zone label per node	→	Graph topology	→	Room type prediction
graph_in input		door / entrance edges		GraphSAGE-Pool output

04 NEXT STEPS

Pipeline

01 **Floor Plan Recreation**

Recreate plan 10000 in Rhino. Room boundaries as closed polylines with shared edges between adjacent rooms — required for TopologicPy to detect adjacency.

02 **Graph Construction — TopologicPy**

Convert the Rhino geometry into a graph. Assign zoning_type to nodes and door/entrance labels to edges to match the MSD input format.

03 **Graph Analysis — NetworkX**

Compute degree, betweenness, and closeness centrality. Analyse spatial connectivity patterns across the floor plan.

04 **Node Classification — GraphSAGE-Pool**

Run the pretrained model on our graph. Predict room_type from zone labels and connectivity. Compare against ground truth from MSD.